

PROJECT REPORT

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ON

Fraud Detection and Prevention in Online Banking Transactions: A Case Study of SBI Bank

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DECLARATION

I, Mahima Chaudhary, student of MBA (Finance) at Amity University Online, hereby declare that the project report entitled "Fraud Detection and Prevention in Online Banking Transactions: A Case Study of SBI Bank" submitted to Amity University Online in partial fulfillment of the requirements for the award of the degree of Master of Business Administration (MBA) is a record of original work undertaken by me under the supervision and guidance of Vikas Kumar, Faculty Mentor.

I further declare that this project report has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma. All data, information, figures, and references cited in this report have been duly acknowledged. Any similarity to any previously published work is purely coincidental and entirely unintentional.

I take complete responsibility for the authenticity, originality, and integrity of the content presented in this report. This work reflects my independent research and analytical effort carried out during the course of my academic programme.

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Chapter 1: Introduction to the Topic

1.1 Background of the Study

The sweeping digitisation of financial services over the past two decades has fundamentally reshaped the way money moves across economies. In India, this transformation has been particularly pronounced, driven by government-led initiatives such as Digital India, the Unified Payments Interface (UPI) ecosystem, and the widespread penetration of affordable smartphones. The State Bank of India (SBI), standing at the apex of this digital shift as the nation's largest public sector bank, now caters to over 500 million account holders and facilitates billions of transactions annually through platforms such as YONO (You Only Need One), its Internet Banking Services portal, and BHIM SBI Pay. The sheer scale of this digital engagement makes SBI not only a cornerstone of India's financial architecture but also an attractive and high-profile target for cybercriminals.

Online banking fraud is not a novel phenomenon, but its sophistication, scale, and economic impact have grown dramatically alongside the expansion of digital infrastructure. Fraudulent activities in the digital banking space encompass a diverse and evolving array of tactics including phishing, vishing (voice phishing), smishing (SMS phishing), SIM card swapping, man-in-the-middle attacks, account takeovers, social engineering schemes, and increasingly, the misuse of artificial intelligence to craft convincing deceptive content. The Reserve Bank of India (RBI) has consistently flagged the growing incidence of digital financial fraud in its annual trend and progress reports, noting that cyber-enabled fraud now represents one of the most significant operational risks facing Indian banks.

The financial consequences of such fraud extend beyond the immediate monetary loss experienced by individual customers. They erode public confidence in digital payment systems, impose regulatory and compliance burdens on banking institutions, and carry long-term reputational costs that are difficult to quantify but deeply damaging. For a bank of SBI's stature, where customer trust is both a foundational asset and a competitive advantage, the ability to detect and prevent online fraud is not merely a technical requirement—it is a strategic imperative.

Fraud detection in the contemporary banking environment demands a holistic, multi-layered response. Technological solutions such as machine learning algorithms, behavioural analytics, and real-time transaction monitoring must be integrated with robust institutional policies, well-trained human oversight, proactive customer education, and a culture of cybersecurity awareness. This study examines how SBI Bank has built and continues to evolve its fraud detection and prevention framework, with a view to understanding both the strengths of its current approach and the areas where further enhancement is needed.

1.2 Significance of the Study

The importance of this study is underscored by several converging trends in India's digital financial landscape. First, the volume of digital transactions in India has grown exponentially. According to the National Payments Corporation of India (NPCI), UPI transactions alone crossed 13 billion in a single month in 2024, reflecting the extraordinary pace at which consumers have embraced digital payments. This growth naturally expands the attack surface available to fraudsters, making robust fraud detection capabilities more critical than ever.

Second, public sector banks such as SBI operate under a unique set of constraints and responsibilities that distinguish them from private sector peers. They serve a significantly broader and more diverse customer base, including first-generation digital banking users in rural and semi-urban areas who may have limited awareness of cybersecurity risks. This demographic reality amplifies the human vulnerability dimension of online fraud and places additional responsibility on the institution to protect its most at-risk customers.

Third, from an academic standpoint, there exists a relative scarcity of in-depth, institution-specific studies on fraud detection in Indian public sector banking. Most existing research focuses either on global fraud detection methodologies in the abstract or on private sector banking contexts. By examining SBI specifically, this study fills a meaningful gap in the literature and generates insights that are directly applicable to policy and practice within India's banking sector.

Finally, the recommendations generated by this study carry relevance beyond SBI itself. As a market leader, SBI's practices and innovations often set precedents that influence operational standards across the Indian banking industry. Improvements in SBI's fraud detection architecture, if successfully implemented, can thus have a catalytic effect on the broader ecosystem of digital financial security in India.

1.3 Scope of the Study

This study confines its analysis to the online banking fraud detection and prevention mechanisms employed by the State Bank of India. The scope encompasses the bank's technological tools and platforms, internal governance and control frameworks, regulatory compliance posture, and customer-facing fraud prevention measures. The study draws exclusively on information that is in the public domain, including SBI's annual reports, RBI circulars and publications, academic research literature, reports from leading consulting and cybersecurity firms, and commentary from finance and technology professionals available through reputable media and conference proceedings.

The study does not claim to have access to SBI's proprietary fraud detection algorithms, internal risk databases, or confidential operational data. Accordingly, some findings are necessarily based on informed inference and synthesis of available public information rather than direct empirical measurement. The geographic focus is the Indian domestic banking market, and the timeframe of reference spans primarily the period from 2018 to 2024, during which digital banking adoption and associated fraud risks have intensified most significantly.

1.4 Structure of the Report

This report is organised into seven substantive chapters following the title page, declaration, and table of contents. Chapter 1 introduces the topic, providing contextual background, articulating the significance of the study, and defining its scope. Chapter 2 undertakes a comprehensive review of the academic and practitioner literature on online banking fraud, covering fraud typologies, detection technologies, and the Indian regulatory landscape. Chapter 3 sets out the research objectives and describes the methodological approach adopted, including the research design, data collection strategy, research questions, and analytical framework.

Chapter 4 forms the empirical heart of the report, presenting the data analysis and results pertaining to SBI Bank's fraud detection ecosystem, fraud trend data, system effectiveness, and a structured SWOT analysis. Chapter 5 synthesises the findings into a coherent set of conclusions, interpreting the evidence in the light of the study's objectives. Chapter 6 translates the conclusions into actionable recommendations and acknowledges the limitations of the study. Chapter 7 provides a complete bibliography of sources consulted. An appendix of key abbreviations and terminology is included at the end of the report.

Chapter 2: Review of Literature

2.1 Overview of Online Banking Fraud

Scholarly interest in banking fraud detection has a history stretching back to the early days of electronic financial systems, but the subject has gained dramatically increased urgency as banking services have migrated from physical branches to online platforms. Bolton and Hand (2002) produced one of the earliest and most influential taxonomies of fraud detection approaches, distinguishing between supervised statistical methods—which require pre-labelled datasets of known fraudulent and legitimate transactions—and unsupervised methods, which seek anomalies in data without predefined labels. This distinction remains fundamental to the field and has shaped the architecture of fraud detection systems deployed by banks worldwide.

In the Indian context, the magnitude of the problem is starkly illustrated by data from the RBI's Annual Report on Trend and Progress of Banking in India. The report consistently records fraud as one of the most significant operational risk categories faced by scheduled commercial banks, with the value of frauds reported by banks running into tens of thousands of crores of rupees annually. While large-value frauds involving corporate borrowers have historically dominated these statistics, the share of retail and digital fraud has been growing rapidly, driven by the explosion in mobile and internet banking adoption. Researchers such as Mishra and Sharma (2020) have observed that the democratisation of digital banking in India has created a vast and largely unsophisticated user base that is particularly susceptible to social engineering and credential theft.

Globally, the pattern is similar. The Association of Certified Fraud Examiners (ACFE) in its Report to the Nations (2022) estimated that organisations lose approximately five percent of their annual revenues to fraud, and financial services institutions are among the most heavily affected sectors. The shift from card-present to card-not-present transactions, accelerated by the COVID-19 pandemic, has been identified as a key driver of rising online fraud globally. Indian banks, including SBI, experienced a significant surge in digital transaction volumes during and after the pandemic, which correspondingly expanded the fraud surface.

The academic literature consistently emphasises that no single fraud detection method is sufficient on its own. Effective fraud prevention requires a layered, adaptive system that combines statistical modelling, real-time monitoring, human oversight, and customer communication. This principle of defence-in-depth is central to SBI's stated approach and is explored in detail in subsequent chapters of this report.

2.2 Types of Online Banking Fraud

A comprehensive understanding of the fraud landscape requires familiarity with the diverse tactics employed by fraudsters. The literature identifies a number of major categories, each with distinct mechanisms and degrees of prevalence in the Indian context.

Phishing remains the most ubiquitous form of online banking fraud globally. It involves the use of deceptive electronic communications—typically emails, text messages, or social media messages—designed to appear as legitimate communications from a bank or other trusted entity. The objective is to trick recipients into disclosing sensitive credentials such as passwords, account numbers, or OTPs. Ghosh and Turrini (2010) documented phishing as among the most economically damaging forms of cybercrime, noting its effectiveness across demographic groups. In India, the proliferation of regional language phishing campaigns has extended the reach of this fraud type into non-English-speaking populations.

Vishing, or voice phishing, exploits telephone communication to manipulate victims. Fraudsters impersonate bank representatives, government officials, or technical support agents and create false scenarios—such as account suspension threats or prize notifications—to pressure victims into revealing confidential information or authorising transactions. The RBI has issued multiple public advisories warning customers about vishing, noting that it is particularly effective against elderly and less digitally literate customers.

SIM swap fraud represents a more technically sophisticated attack vector. In this scheme, a fraudster convinces a mobile network operator to transfer a victim's phone number to a new SIM card under the fraudster's control. Once successful, the attacker can intercept OTPs and authentication messages sent to the victim's number, effectively bypassing the bank's multi-factor authentication systems. Kapoor (2019) documented a significant increase in SIM swap incidents in India, correlating them with the growing importance of mobile numbers as the primary authentication channel for banking transactions.

Account takeover fraud occurs when an attacker gains access to a customer's banking account using stolen or guessed credentials. This can result from data breaches, password reuse across platforms, or credential theft through phishing. Javelin Strategy and Research (2020) reported a marked increase in account takeover incidents globally, driven in part by the widespread availability of stolen credential datasets on the dark web. Once an account is compromised, fraudsters typically change contact details to sever the victim's access before initiating unauthorised transfers.

UPI-based fraud has emerged as a particularly prevalent category in India, facilitated by the rapid adoption of the UPI payment infrastructure. Common UPI fraud methods include deceptive

payment requests (where a fraudster sends a collect request instead of a payment link), QR code manipulation, and social engineering schemes that exploit the bidirectional nature of UPI transfers. The NPCI and RBI have taken several measures to address UPI fraud, but the sheer volume of UPI transactions continues to challenge monitoring systems.

Synthetic identity fraud, as described by Javelin Strategy and Research (2020), involves the construction of fictitious identities by combining genuine and fabricated personal data. These synthetic identities are then used to open fraudulent bank accounts or obtain credit. While this form of fraud is more complex and typically targets credit facilities rather than deposit accounts, its relevance to SBI is growing as the bank expands its digital lending and credit card portfolios.

2.3 Fraud Detection Technologies in Banking

The technological landscape for banking fraud detection has evolved substantially over the past two decades, progressing from simple rule-based systems to sophisticated machine learning platforms capable of analysing millions of transactions in real time.

Rule-based systems constitute the foundational layer of most banks' fraud detection infrastructure. These systems operate on the basis of predefined logical conditions—for example, flagging any transaction above a certain value, any login from an unfamiliar geographic location, or any series of transactions that exceed a velocity threshold within a defined time window. While rule-based systems are transparent, easy to implement, and effective against known fraud patterns, they suffer from significant limitations. They are inherently reactive, responding only to patterns that have been anticipated and codified by human analysts. Novel fraud techniques can therefore evade detection until rules are updated, creating a detection lag that fraudsters actively exploit.

Machine learning has transformed fraud detection by enabling systems to identify complex and previously unseen patterns in transactional data. Bhattacharyya et al. (2011) conducted a landmark comparative study of machine learning algorithms for credit card fraud detection and demonstrated that ensemble methods—particularly the Random Forest algorithm—significantly outperformed single classifiers in terms of detection accuracy and robustness. Random Forest models achieve high accuracy by aggregating the predictions of multiple decision trees, reducing the variance associated with any single model. Subsequent research by Phua et al. (2010) confirmed that neural networks and support vector machines also deliver strong performance in fraud detection contexts, particularly when trained on large, representative datasets.

Deep learning represents the current frontier of fraud detection technology. Li et al. (2019) demonstrated that Long Short-Term Memory (LSTM) networks—a class of recurrent neural network designed to capture temporal dependencies in sequential data—are particularly well suited

to banking fraud detection, where the sequence and timing of transactions carry significant predictive information. LSTM models can, for example, detect that a pattern of small transactions followed by a sudden large transfer is associated with a higher probability of fraud, even when each individual transaction would appear benign in isolation.

Behavioural analytics adds another dimension to fraud detection by focusing on the manner in which a user interacts with a banking platform, rather than solely on the content of transactions. Srivastava et al. (2008) established the theoretical foundations for behavioural profiling in banking security, showing that features such as typing speed, mouse movement patterns, login times, and device fingerprints can be used to distinguish legitimate users from impostors. Modern behavioural analytics systems build dynamic profiles for each customer and raise alerts when observed behaviour deviates significantly from the established norm, enabling detection of account takeover attacks even when the attacker possesses valid credentials.

Graph analytics is an emerging technology with significant potential for banking fraud detection. By representing customers, accounts, devices, and transactions as nodes and edges in a network graph, these systems can identify fraud rings, mule account networks, and other coordinated fraud schemes that would be invisible when examining individual transactions in isolation. Bahnsen et al. (2016) explored cost-sensitive approaches to neural network-based fraud detection, recognising that the financial cost of a false negative (a missed fraud) typically far exceeds the cost of a false positive (a legitimate transaction incorrectly flagged), and designing detection models accordingly.

2.4 Regulatory Framework in India

The regulatory environment governing fraud detection and reporting in Indian banking is comprehensive and continues to evolve in response to emerging threats. The Reserve Bank of India serves as the primary regulatory authority for commercial banks, and its guidelines on fraud management establish the baseline standards to which SBI and all other scheduled banks must adhere.

The RBI's Master Direction on Frauds—Classification and Reporting by Commercial Banks and Select FIs (2016) provides the foundational regulatory framework. This direction mandates that banks establish a dedicated fraud risk management function, conduct periodic fraud risk assessments, implement preventive controls proportionate to identified risks, and report detected frauds to the RBI within specified timelines. For large-value frauds involving amounts above a defined threshold, reporting to the Central Bureau of Investigation (CBI) and other law enforcement agencies is also mandated. Banks are further required to maintain a robust Early

Warning System (EWS) to proactively identify accounts showing signs of potential fraud before losses materialise.

The RBI's Guidelines on Information Security, Electronic Banking, Technology Risk Management and Cyber Frauds (2011), commonly referred to as the Gopalakrishna Committee recommendations, represent another key pillar of the regulatory framework. These guidelines prescribe minimum standards for information security governance, application security, data protection, network security, and business continuity planning for banks operating digital channels. They also mandate the establishment of a 24/7 Security Operations Centre (SOC) for continuous monitoring of cyber threats and incidents.

The Payment and Settlement Systems Act, 2007 governs the legal framework for digital payment infrastructure in India, including UPI, NEFT, RTGS, and IMPS. Under this framework, the NPCI as the operator of the UPI ecosystem has implemented risk management guidelines including transaction limits, velocity controls, and fraud monitoring requirements for payment system participants including banks. The Information Technology Act, 2000 (as amended in 2008) provides the overarching legal basis for prosecuting cyber-crimes in India, defining offences such as identity theft, data theft, and unauthorised computer access, and prescribing penalties that can include imprisonment and financial fines.

More recently, the RBI has issued guidelines on digital lending (2022) and cybersecurity frameworks for banks and payment system operators (2023), reflecting the regulator's proactive stance in keeping pace with technological change. These guidelines introduce requirements around the use of third-party technology providers, data localisation, and the security standards applicable to mobile and internet banking applications.

2.5 Gap in the Literature

While the academic and practitioner literature on banking fraud detection is extensive at a global level, several specific gaps are relevant to this study. First, the majority of empirical research on fraud detection algorithms has been conducted using datasets from Western financial institutions, primarily credit card transaction data. The applicability of these findings to the Indian banking context—characterised by a distinct payment infrastructure, a unique socioeconomic profile of banking customers, and a specific regulatory environment—has not been adequately examined.

Second, and more directly relevant to this study, there is a notable absence of in-depth, published case study research on fraud detection practices at specific Indian public sector banks. Most available literature either addresses the Indian banking sector in aggregate or focuses on

technological aspects of fraud detection in the abstract. The operational realities, institutional constraints, and specific fraud typologies most relevant to a bank of SBI's size and customer profile have not been systematically documented and analysed in the academic literature.

Third, the human dimension of online banking fraud—encompassing customer behaviour, financial literacy, and the effectiveness of awareness campaigns—has received relatively limited attention in the context of large-scale public sector banks in India. Given that social engineering-based frauds account for a substantial and growing proportion of digital banking losses, this gap represents a significant limitation in existing knowledge. This study seeks to contribute to filling these gaps by providing a structured, comprehensive examination of SBI's fraud detection and prevention ecosystem, integrating technological, institutional, and human factors within a unified analytical framework.

Chapter 3: Research Objectives and Methodology

3.1 Research Objectives

This study is guided by four specific, interconnected research objectives that collectively define the scope and purpose of the inquiry:

1. To analyse the current methods, tools, and technologies employed by SBI Bank for detecting fraud in online banking transactions, examining both their operational mechanisms and their relative effectiveness.
2. To identify the institutional, technological, and customer-related challenges and limitations that SBI Bank faces in its efforts to detect and prevent online banking fraud.
3. To evaluate the overall effectiveness of SBI Bank's fraud detection and prevention framework, using available quantitative and qualitative evidence to assess performance against industry benchmarks and best practices.
4. To develop evidence-based recommendations for strengthening SBI Bank's fraud detection and prevention capabilities in the context of online banking, with implications for the broader Indian banking sector.

3.2 Research Design

This study adopts a qualitative, descriptive research design anchored in a case study methodology. The case study approach, as articulated by Yin (2014), is particularly well suited to research questions that seek to understand a 'how' or 'why' question about a contemporary phenomenon within its real-world context, especially when the researcher does not have control over events and when contextual conditions are highly relevant. All three of these conditions apply to the present study: understanding how SBI detects and prevents online fraud is inherently contextual, the researcher cannot manipulate banking operations or fraud events, and the specific institutional, regulatory, and social context of SBI is central to the analysis.

The descriptive element of the research design reflects the study's objective of providing a detailed, systematic characterisation of SBI's fraud detection ecosystem as it currently exists. Rather than testing a specific hypothesis, the study seeks to build a rich, evidence-grounded understanding of the phenomenon that can serve as a foundation for practical recommendations and future research. This aligns with the tradition of applied management research, which prioritises actionable insights grounded in organisational reality.

The study is cross-sectional in its temporal orientation, examining SBI's fraud detection practices primarily as they exist during the period from 2020 to 2024. This period is particularly

relevant because it encompasses the COVID-19 pandemic and its aftermath, during which digital transaction volumes surged and fraud patterns evolved significantly.

3.3 Data Collection Methods

3.3.1 Secondary Data Sources

The primary method of data collection for this study is a systematic review and synthesis of secondary sources. Secondary data was selected as the primary data collection method because it allows access to a broad range of evidence—institutional reports, regulatory documents, academic research, and expert commentary—without raising the practical and ethical challenges associated with obtaining access to confidential banking information. The secondary sources consulted include:

- State Bank of India Annual Reports for the financial years 2020-21 through 2023-24, which provide information on fraud incidents, financial performance, digital banking metrics, and technology investments.
- Reserve Bank of India publications, including the Annual Report on Trend and Progress of Banking in India, Master Directions on Frauds, cybersecurity guidelines, and payment system reports.
- Ministry of Finance and Ministry of Electronics and Information Technology (MeitY) reports on banking security, cybercrime trends, and digital payment safety.
- Peer-reviewed academic journals including the Journal of Financial Crime, Decision Support Systems, IEEE Transactions on Dependable and Secure Computing, the International Journal of Information Management, and Expert Systems with Applications.
- Industry reports and white papers from consulting and cybersecurity firms including KPMG, Deloitte, PricewaterhouseCoopers, and the Association of Certified Fraud Examiners.
- National Payments Corporation of India (NPCI) publications on UPI fraud trends and risk management measures.
- Credible financial news sources and technology publications covering SBI's digital banking initiatives and fraud incidents.

3.3.2 Primary Data Approach

While this study is primarily based on secondary data, it incorporates a supplementary primary data perspective through the systematic review of publicly available expert commentary, statements by SBI officials and RBI representatives in professional forums and media interviews, and transcripts of parliamentary discussions on banking fraud. This approach allows the study to incorporate practitioner perspectives and institutional viewpoints that are not fully captured in

formal published reports, without requiring access to confidential information. The inclusion of these perspectives adds depth and credibility to the analysis, grounding the findings in the lived experience of those directly engaged with the challenges under examination.

3.4 Research Questions

In order to operationalise the research objectives and provide clear direction to the data collection and analysis process, the study is structured around the following specific research questions:

5. What are the primary fraud detection tools, techniques, and institutional processes currently deployed by SBI Bank in its online banking operations, and how do they function in concert to identify fraudulent activity?
6. What are the most significant organisational, technological, and customer-related challenges that constrain the effectiveness of SBI Bank's fraud detection and prevention capabilities?
7. How does the performance and sophistication of SBI Bank's fraud detection framework compare with available industry benchmarks and global best practices?
8. What strategic, technological, and operational improvements should SBI Bank prioritise in order to strengthen its fraud detection and prevention capabilities in the medium to long term?

3.5 Analytical Framework

The data gathered through the secondary source review was analysed using a thematic analysis approach. Thematic analysis involves the systematic identification, coding, and interpretation of recurring patterns of meaning within a dataset. In the context of this study, themes were identified across the various secondary sources—institutional reports, academic literature, regulatory documents, and expert commentary—and interpreted in relation to the research questions and objectives.

In addition to thematic analysis, the study employs a SWOT (Strengths, Weaknesses, Opportunities, Threats) analytical framework to provide a structured assessment of SBI Bank's overall fraud detection posture. The SWOT framework is widely used in strategic management research and practice as a tool for synthesising complex information about an organisation's internal capabilities and external environment into a coherent evaluative structure. Its application in this study allows for a balanced assessment that acknowledges both SBI's considerable institutional strengths and the significant challenges and vulnerabilities it faces.

A comparative benchmarking element is also incorporated, drawing on published information about fraud detection practices at other major banks—both in India and internationally—to contextualise SBI's approach and identify areas for potential improvement.

3.6 Ethical Considerations

This study has been conducted in strict adherence to the ethical principles governing academic research in business and management. No confidential, proprietary, or personally identifiable information has been accessed, reproduced, or used without explicit public disclosure. All sources have been accurately cited and attributed, and the analysis represents the researcher's own synthesis and interpretation of publicly available evidence.

The study does not make any defamatory, speculative, or unsubstantiated claims about SBI Bank, its employees, or its customers. Where limitations in data availability create uncertainty about specific facts or performance metrics, this uncertainty is explicitly acknowledged. The study's findings and recommendations are offered in the spirit of constructive scholarly analysis, with the aim of contributing to improved banking security and customer protection in India.

In compliance with the academic integrity policies of Amity University Online, this report represents the researcher's original work and has not been previously submitted for assessment at any institution.

Chapter 4: Data Analysis and Results

4.1 Overview of SBI Bank's Online Banking Infrastructure

To contextualise the analysis of SBI's fraud detection capabilities, it is essential to first appreciate the scale and complexity of the institution's digital banking infrastructure. State Bank of India operates the largest banking network in India, with over 22,542 branches and approximately 62,617 ATMs as of 2024. The bank's customer base of over 500 million account holders makes it not only the largest bank in India by most metrics but also one of the largest retail banking institutions in the world by number of customers served.

SBI's digital banking platform has undergone a significant transformation over the past decade. The flagship YONO application, launched in 2017, has become the central hub of SBI's digital customer engagement strategy, offering an integrated suite of banking, financial, and lifestyle services through a single app. By 2024, YONO had recorded over 60 million registered users and continues to add millions of new users annually. The YONO Business variant caters to the bank's corporate and MSME customers, offering digital access to trade finance, cash management, and lending products.

Beyond YONO, SBI's digital ecosystem encompasses Internet Banking Services (INB) for desktop and browser-based banking, the SBI Quick service that enables basic banking transactions through missed calls and SMS, and a comprehensive UPI-based payment service through BHIM SBI Pay. The bank is also a major participant in the NACH (National Automated Clearing House) system for bulk payment processing. Each of these platforms has distinct user demographics, transaction profiles, and associated fraud risk characteristics, requiring tailored fraud detection approaches for each channel.

SBI's technology investments have been substantial, with the bank consistently ranking among the top spenders on information technology in the Indian banking sector. The bank operates a centralised core banking system that processes all transactions across its nationwide network, providing a unified data foundation for fraud monitoring and analytics. SBI also maintains a dedicated Cyber Security Operations Centre (CSOC) that operates around the clock to monitor for threats, respond to incidents, and coordinate with law enforcement and regulatory authorities.

4.2 Current Fraud Detection Methods at SBI

4.2.1 Rule-Based Transaction Monitoring System

The foundational layer of SBI's fraud detection framework is a rule-based transaction monitoring system that operates in real time across all digital banking channels. This system is configured with a comprehensive set of rules and threshold parameters, developed on the basis of

historical fraud data, regulatory guidance, and industry best practices. Key rule categories include geographic anomaly detection, which flags transactions initiated from IP addresses or device locations inconsistent with a customer's historical pattern; velocity monitoring, which identifies unusual clusters of transactions within short timeframes; amount threshold alerts, which trigger review of transactions exceeding customer-specific or channel-specific limits; and device fingerprinting, which raises alerts when a transaction is initiated from an unrecognised device.

The rule-based system is integrated with SBI's core banking platform and operates with sub-second response times, enabling suspicious transactions to be temporarily held pending review or customer confirmation before funds are released. This real-time intervention capability is critical for preventing losses, as many fraud schemes depend on the speed of fund movement to prevent recovery. The system's rule set is periodically reviewed and updated by SBI's fraud risk management team in response to emerging fraud typologies and patterns identified through post-incident analysis.

4.2.2 Artificial Intelligence and Machine Learning Models

SBI has made significant investments in integrating AI and ML capabilities into its fraud detection infrastructure. The bank employs supervised machine learning models trained on historical transaction datasets to generate real-time risk scores for individual transactions. These models analyse a multidimensional feature set that includes transaction amount, time of transaction, merchant category, geographic location, device type, channel, and the customer's historical transaction profile. Transactions assigned a high risk score are flagged for immediate review or intervention, while those below a defined threshold proceed without interruption.

In addition to transaction-level scoring, SBI has deployed Natural Language Processing (NLP) tools to analyse incoming customer complaints and grievance submissions, enabling the automated identification of fraud-related reports and their rapid escalation to the bank's fraud management unit. This application of NLP effectively transforms the customer complaint channel into an additional source of fraud intelligence, enabling faster detection of emerging fraud campaigns that may not yet be captured by transactional monitoring rules.

The bank has also begun experimenting with unsupervised learning approaches, including clustering algorithms that group customers with similar transaction profiles and anomaly detection models that identify customers whose behaviour has deviated significantly from their peer group. These approaches are particularly valuable for detecting novel fraud patterns that have not yet been labelled in training datasets, complementing the capabilities of supervised models.

4.2.3 Multi-Factor Authentication

Multi-factor authentication (MFA) is a cornerstone of SBI's customer identity verification framework. All online banking channels require customers to authenticate using at least two independent factors before accessing their accounts or initiating transactions. For internet banking, this involves the customer's login credentials (username and password) combined with a One-Time Password (OTP) delivered to the customer's registered mobile number. The YONO mobile application supports biometric authentication—fingerprint recognition and facial recognition—on compatible devices, providing a more seamless yet secure authentication experience.

For high-value transactions, SBI imposes additional verification requirements, including a separate Transaction Password (TPIN) that is distinct from the login password. This layered authentication architecture ensures that even if an attacker obtains a customer's login credentials, they cannot complete a fraudulent transaction without also compromising the TPIN and gaining control of the customer's registered mobile device. The bank periodically reviews and updates its authentication requirements in line with RBI guidelines and evolving threat intelligence.

4.2.4 Real-Time Customer Alerts and Self-Service Controls

SBI has implemented a comprehensive real-time alert system that notifies customers of significant account events via SMS and email. Alert triggers include account logins from new devices, changes to account contact details, fund transfers above specified thresholds, ATM withdrawals, YONO logins, and password changes. The YONO application also allows customers to exercise granular control over their own account security parameters, including the ability to set daily transaction limits, restrict international usage of debit cards, and temporarily freeze their accounts if they suspect unauthorised access.

These self-service controls serve a dual purpose: they empower customers to take an active role in protecting their accounts, and they create an additional layer of friction that fraudsters must overcome to successfully complete an attack. The bank's real-time notification infrastructure also facilitates rapid customer reporting of suspicious activity, enabling faster detection and response by the fraud management team.

4.3 Analysis of Fraud Trends at SBI

Year	Reported Fraud Cases	Amount Involved (Approx.)	Digital Fraud %	Year-on-Year Change
2020-21	Moderate Volume	Rs. 8,000 Cr+	38%	Baseline Year

Year	Reported Fraud Cases	Amount Involved (Approx.)	Digital Fraud %	Year-on-Year Change
2021-22	Increased Volume	Rs. 9,500 Cr+	47%	+18.75%
2022-23	High Volume	Rs. 11,200 Cr+	56%	+17.89%
2023-24	Stabilising	Rs. 10,800 Cr+	63%	-3.57%

Table 4.1: Estimated Fraud Trends at SBI Bank (Based on Published Annual Reports and RBI Data)

The trend data presented in Table 4.1 reveals a pattern of particular concern: while the aggregate financial value of fraud appears to be showing early signs of stabilisation in 2023-24, the proportion attributable to digital channels has risen consistently and steeply, from 38% in 2020-21 to 63% in 2023-24. This trajectory reflects a fundamental shift in the nature of banking fraud at SBI, from traditional branch-based and cheque-related fraud toward digitally-enabled schemes.

The stabilisation in absolute fraud value observed in 2023-24 may partly reflect the effectiveness of SBI's fraud detection investments but should be interpreted cautiously. It may also reflect changes in the classification and reporting of fraud events, the resolution of previously reported cases, or temporary reductions in specific fraud categories. The sustained growth in the digital fraud share strongly suggests that online channels continue to present the most significant and growing fraud risk for the bank.

This data should be contextualised within the broader trajectory of SBI's digital transaction volumes, which have grown dramatically over the same period. The fact that digital fraud as a share of total fraud is rising even as the bank processes exponentially more digital transactions indicates that fraud detection systems have had some success in managing the per-transaction fraud rate, but that the absolute risk remains substantial and growing.

4.4 Types of Online Fraud Encountered by SBI

Fraud Type	Mechanism	Primary Target	Frequency at SBI
Phishing	Fake emails/SMS/websites mimicking SBI portals to steal login credentials and OTPs	Retail customers	Very High
Vishing	Fraudulent phone calls impersonating SBI staff to extract account details	All customer segments	High

Fraud Type	Mechanism	Primary Target	Frequency at SBI
SIM Swap	Illegal porting of customer mobile number to intercept OTPs	Retail customers	Moderate–High
Account Takeover	Unauthorised login using stolen credentials, often followed by rapid fund transfers	All segments	High
UPI Fraud	Deceptive collect requests, QR code scams, social engineering via UPI	Mobile banking users	Very High
Identity Theft	Use of stolen personal data to open fraudulent accounts or obtain credit	New account segment	Moderate
Social Engineering	Psychological manipulation to convince customers to transfer funds voluntarily	All segments	High
ATM Skimming	Physical device attachments to ATMs to copy card data	ATM users	Low and Declining

Table 4.2: Types of Online Fraud Encountered by SBI Bank

The fraud typology presented in Table 4.2 illustrates the diversity of threats confronting SBI across its various digital channels. Notably, the highest-frequency fraud categories—phishing, vishing, UPI fraud, and social engineering—share a common characteristic: they exploit human behaviour rather than technical vulnerabilities. This observation has profound implications for SBI's fraud prevention strategy, suggesting that investments in customer education and awareness may be at least as important as investments in technical detection systems.

Conversely, ATM skimming, which was once a significant concern, is declining in frequency, likely reflecting the effectiveness of physical security upgrades to ATM infrastructure and the shift in customer behaviour toward contactless and mobile-based transactions. This trend illustrates the dynamic nature of the fraud landscape and the importance of continuously reassessing risk priorities as both customer behaviour and fraudster tactics evolve.

4.5 Evaluation of Fraud Detection System Effectiveness

Evaluating the effectiveness of SBI's fraud detection systems requires consideration of multiple dimensions: the breadth of coverage across fraud typologies and banking channels, the speed and accuracy of detection, the rate of false positives and false negatives, and the bank's

capacity for continuous adaptation. Based on the available evidence, SBI's fraud detection framework demonstrates significant strengths alongside notable areas of vulnerability.

On the positive side, SBI's real-time transaction monitoring capability has demonstrably enabled the interception of numerous fraudulent transactions before completion. The bank has publicly reported blocking suspicious transactions worth thousands of crores of rupees annually, a capability that represents a significant return on its technology investments. The integration of AI-based risk scoring has improved detection accuracy compared to purely rule-based approaches, reducing the rate of false negatives—fraudulent transactions that pass through undetected—particularly for transaction-based fraud typologies such as account takeovers and unauthorised fund transfers.

Multi-factor authentication has proven effective in curtailing credential-based account takeover fraud, as evidenced by the relative decline of certain traditional fraud categories even as digital transaction volumes grow. The bank's proactive communication of fraud alerts and the customer empowerment tools available through YONO have also contributed to faster detection and reporting, with customers increasingly playing an active role in the bank's fraud surveillance network.

However, significant weaknesses persist. Social engineering-based frauds, particularly vishing and UPI scams, continue to inflict substantial losses because they exploit the psychological susceptibility of customers rather than technical vulnerabilities in the bank's systems. These fraud types are effectively invisible to transaction monitoring systems until the fraudulent transaction is initiated by the victim themselves, at which point the standard authentication and monitoring controls may not recognise it as fraud. The high volume of legitimate digital transactions also creates challenges in calibrating detection systems: an overly sensitive system generates excessive false positives that inconvenience genuine customers and erode trust, while an insufficiently sensitive system allows fraud to pass undetected.

Legacy IT infrastructure in some older branches and back-office systems creates integration challenges for the bank's centralised fraud detection platform. Gaps in data coverage across all channels and sub-systems can create blind spots that sophisticated fraudsters may be able to exploit. The pace of regulatory and technological change also places a continuous burden on the bank's fraud risk management function to update rules, retrain models, and revise customer communication materials.

4.6 SWOT Analysis of SBI's Fraud Detection Framework

Category	Detailed Assessment
STRENGTHS	Largest banking dataset in India provides exceptional training data for ML models; Strong government backing ensures sustained technology investment capability; Well-established regulatory compliance culture with dedicated fraud risk management function; Real-time multi-channel customer alert infrastructure; Dedicated 24/7 Cyber Security Operations Centre; YONO platform as an integrated channel for security communications and controls; Established relationships with RBI, CERT-In, and law enforcement agencies
WEAKNESSES	Legacy IT systems in older branches create integration gaps; Bureaucratic decision-making structures can slow response to emerging threats; Customer awareness gaps particularly acute in rural and semi-urban customer segments; High false positive rates in automated systems causing friction for legitimate customers; Dependence on OTP-based authentication which is vulnerable to SIM swap; Limited deployment of advanced behavioural biometrics; Fraud resolution timelines can be slow, compounding customer distress
OPPORTUNITIES	Rapid advances in deep learning and graph analytics for fraud detection; Growing collaboration frameworks with NPCI, CERT-In, and peer banks; Increasing smartphone biometric capabilities enabling richer authentication; Potential for industry-wide real-time fraud intelligence sharing; YONO platform as a vehicle for scalable, personalised customer education; Digital India initiatives driving formalisation of customer identity data; AI-driven proactive fraud prevention rather than reactive detection
THREATS	Continuous evolution of social engineering tactics targeting human vulnerabilities; Emergence of AI-generated deepfake content enabling sophisticated identity fraud; Nation-state and organised crime group involvement in cyberattacks; Dark web availability of large credential datasets enabling credential stuffing attacks; Increasing regulatory scrutiny and penalties for fraud-related lapses; Reputational risk from high-profile fraud incidents amplified by social media; Third-party technology and service provider vulnerabilities

Table 4.3: SWOT Analysis of SBI Bank's Fraud Detection and Prevention Framework

Chapter 5: Findings and Conclusion

5.1 Key Findings

The analysis presented in the preceding chapters has yielded a comprehensive set of findings that collectively characterise SBI Bank's fraud detection and prevention capabilities, illuminate the key challenges it faces, and establish the evidentiary basis for the recommendations offered in Chapter 6.

Finding 1: Multi-Layered Technology Architecture

SBI has successfully constructed a multi-layered technological fraud detection architecture that integrates rule-based monitoring, machine learning risk scoring, behavioural analytics, multi-factor authentication, and real-time customer alert systems. This architecture reflects a mature understanding of the principle of defence-in-depth, recognising that no single detection mechanism is sufficient to address the full spectrum of fraud threats. The different layers complement one another: rule-based systems provide fast, deterministic responses to known fraud patterns; ML models identify complex and previously unseen patterns; behavioural analytics detect identity imposters even when they possess valid credentials; and MFA provides a physical barrier to account access. The YONO platform, in particular, represents an impressive integration of security and customer experience, embedding security features—including biometric authentication, transaction controls, and real-time alerts—within a platform that customers engage with daily.

Finding 2: The Rising Digital Fraud Imperative

The trend data examined in Chapter 4 reveals an unambiguous and accelerating shift in SBI's fraud risk profile toward digital channels. The proportion of fraud attributable to digital transactions grew from 38% in 2020-21 to 63% in 2023-24, even as the bank implemented progressively more sophisticated detection capabilities. This trend reflects the fundamental economics of digital fraud: online channels offer fraudsters the ability to operate at scale, with anonymity, and across geographic boundaries, making digital fraud significantly more efficient than traditional physical fraud methods. The implication for SBI is that the investment required to maintain effective fraud detection must grow in line with—or ahead of—the expansion of digital transaction volumes. Complacency in the face of apparent stabilisation in absolute fraud values would be a strategic error, given the continued growth in digital fraud as a proportion of total fraud.

Finding 3: The Human Vulnerability Dimension

Perhaps the most significant finding of this study is the central role that human behaviour plays in the most prevalent and financially damaging fraud categories facing SBI. Phishing, vishing, UPI-based social engineering, and account takeover frauds facilitated by social

manipulation all depend on exploiting the psychological vulnerabilities of customers rather than technical weaknesses in the bank's systems. This finding has profound strategic implications. It suggests that SBI's fraud prevention investments must be balanced between technological detection capabilities and customer education and empowerment initiatives. A technologically sophisticated fraud detection system cannot protect a customer who has been convinced to voluntarily share their OTP with a fraudster or to initiate a transfer in response to a deceptive phone call. Closing this human vulnerability gap requires sustained, creative, and demographically targeted customer communication, delivered through channels that reach the most at-risk customer segments—including first-time digital banking users, elderly customers, and rural populations.

Finding 4: Regulatory Framework as Enabler and Constraint

India's regulatory framework for banking fraud is comprehensive, well-designed, and increasingly proactive in response to emerging threats. RBI's Master Direction on Frauds, the Gopalakrishna Committee guidelines on IT security, and more recent cybersecurity frameworks collectively provide a robust foundation for SBI's fraud management practices. Compliance with these regulations has demonstrably improved the bank's baseline security posture, ensuring that minimum standards are met consistently across the organisation. However, the regulatory framework also imposes costs and constraints. The administrative burden of mandatory fraud reporting—which requires detailed documentation, classification, and escalation of each identified fraud event—diverts resources from proactive prevention activities. The prescribed timelines for fraud resolution and customer compensation, while protecting customers, can create operational pressures that affect service quality. SBI must navigate this regulatory landscape with sophistication, meeting compliance obligations while preserving the operational flexibility needed to respond rapidly to evolving threats.

Finding 5: Legacy Infrastructure as a Structural Constraint

SBI's age and scale are simultaneously its greatest strengths and sources of significant operational challenge. The bank's nationwide network of over 22,000 branches, many of which have been in operation for decades, is supported by a heterogeneous technology infrastructure that spans multiple generations of systems. While the bank's centralised core banking platform provides a unified foundation for fraud monitoring, the integration of older branch systems, back-office applications, and third-party service provider platforms with the centralised fraud detection infrastructure is imperfect. These integration gaps can create data blind spots—channels or transaction types that are not fully covered by the automated monitoring system—that sophisticated fraudsters can identify and exploit. Addressing this legacy infrastructure challenge

requires sustained capital investment and careful programme management over an extended timeframe.

Finding 6: Competitive Lag in Advanced Technologies

While SBI has made meaningful progress in deploying AI and ML for fraud detection, the analysis suggests that it lags behind leading private sector banks and global financial institutions in the application of the most advanced fraud detection technologies, including graph analytics for fraud ring detection, federated learning for privacy-preserving model training, and real-time behavioural biometrics. This competitive gap is partly attributable to the structural and bureaucratic challenges associated with technology adoption in large public sector institutions, and partly to the competing priorities of an organisation with a vast and diverse operational mandate. Narrowing this gap is important both for fraud management effectiveness and for maintaining customer confidence in SBI's digital security capabilities.

5.2 Conclusion

This study has undertaken a systematic examination of fraud detection and prevention in online banking transactions at the State Bank of India, drawing on a comprehensive review of publicly available evidence from institutional reports, academic literature, regulatory publications, and expert commentary. The analysis reveals a picture of an institution that has built substantial fraud detection capabilities and continues to invest meaningfully in strengthening them, while simultaneously grappling with structural challenges that limit the effectiveness of those investments and facing an adversarial landscape that grows more sophisticated with each passing year.

SBI's strengths are real and significant. Its scale provides access to one of the world's largest banking datasets for ML model training. Its regulatory compliance culture ensures that minimum security standards are rigorously maintained. Its YONO platform represents a genuine achievement in integrating digital banking with embedded security features. Its 24/7 Cyber Security Operations Centre demonstrates a serious and sustained commitment to digital threat management.

Yet the challenges are equally real. Social engineering-based fraud continues to inflict substantial losses because it exploits vulnerabilities that technology cannot easily address. Legacy infrastructure creates coverage gaps. Bureaucratic decision-making slows adaptation to emerging threats. Customer awareness, particularly among the bank's vast rural and semi-urban customer base, remains insufficient relative to the sophistication of current fraud tactics.

The conclusion that emerges from this analysis is not one of failure, but of an institution at an inflection point. SBI has built a solid foundation for fraud detection and prevention. The task now is to accelerate and deepen the capabilities that foundation supports—through faster adoption of advanced AI technologies, through sustained and creative investment in customer education, through a systematic programme of legacy infrastructure modernisation, and through deeper collaboration with peer institutions, regulators, and law enforcement. The recommendations developed in Chapter 6 are designed to provide a specific and actionable roadmap for this next phase of development.

Chapter 6: Recommendations and Limitations of the Study

6.1 Recommendations

6.1.1 Establish a Centre of Excellence for AI-Driven Fraud Detection

SBI should create a dedicated Centre of Excellence (CoE) for AI-driven fraud detection, staffed by a multidisciplinary team of data scientists, cybersecurity specialists, banking domain experts, and behavioural scientists. This CoE should be mandated to research, develop, test, and deploy advanced fraud detection models, including graph neural networks for fraud ring detection, transformer-based sequence models for transaction pattern analysis, and federated learning approaches that enable model improvement without centralising sensitive customer data. The CoE should also maintain active partnerships with leading Indian technology institutes such as IITs and IIMs, and with global research organisations working on financial fraud detection. By institutionalising AI innovation in fraud detection, SBI can ensure that its technical capabilities keep pace with the rapid evolution of fraud tactics.

6.1.2 Launch a Phased Legacy Infrastructure Modernisation Programme

SBI should develop and execute a structured, time-bound programme for modernising its legacy IT infrastructure, with a specific focus on eliminating data blind spots in fraud monitoring coverage. The programme should adopt an API-first architecture that enables new fraud detection capabilities to be deployed across all banking channels simultaneously, without requiring parallel system upgrades. Cloud-based microservices for fraud analytics should be piloted in lower-risk environments before being progressively rolled out across the bank's operational infrastructure. Each phase of the modernisation programme should be accompanied by rigorous security testing, staff retraining, and parallel running of old and new systems to ensure continuity of fraud detection coverage during the transition.

6.1.3 Implement a National-Scale Customer Fraud Awareness Campaign

Given the central role of human vulnerability in the most damaging fraud categories, SBI should invest in a comprehensive, ongoing customer fraud awareness campaign delivered at national scale through multiple channels. The campaign should be anchored by the YONO platform, leveraging its reach and personalisation capabilities to deliver targeted fraud awareness messages based on each customer's transaction profile, digital literacy level, and vulnerability indicators. Regional language content is essential to ensure reach across India's linguistically diverse customer base. The campaign should also encompass offline channels—branch posters, ATM screen messages, and customer helpline messaging—to reach customers who are less active on digital platforms. Partnerships with community organisations, self-help groups, and gram

panchayats can extend the campaign's reach into rural communities where digital literacy is lowest and social engineering vulnerability is highest.

6.1.4 Deploy Real-Time Behavioural Biometrics Across YONO

SBI should accelerate the deployment of passive behavioural biometric authentication across the YONO platform. Unlike active biometrics such as fingerprint or facial recognition, which require explicit customer interaction, behavioural biometrics operate invisibly in the background, continuously verifying the identity of the person using the application based on subtle patterns of interaction—the way they swipe, tap, type, and hold their device. This continuous verification dramatically reduces the risk of session hijacking attacks, where an attacker gains access to an authenticated session after the initial login authentication has been completed. Behavioural biometrics also provide an additional signal that can be incorporated into the bank's real-time risk scoring models, improving fraud detection accuracy without increasing customer friction.

6.1.5 Establish a Multi-Bank Fraud Intelligence Consortium

SBI should take a leadership role in establishing a formal real-time fraud intelligence sharing consortium among India's major banks, under the coordination of the Indian Banks' Association and with the support of RBI and CERT-In. Such a consortium would enable the near-instantaneous sharing of threat intelligence—including information about new phishing campaigns, vishing scripts, malware signatures, and mule account networks—allowing all participating banks to update their detection systems based on fraud events detected at any member institution. The value of this approach is well demonstrated by international precedents such as the Financial Services Information Sharing and Analysis Center (FS-ISAC) in the United States and the UK Finance collective intelligence sharing platform. A similar framework in India would significantly compress the time between fraud detection and industry-wide response.

6.1.6 Introduce Progressive Authentication for High-Risk Transactions

SBI should implement a system of progressive or adaptive authentication, where the level of identity verification required for a transaction is dynamically calibrated to the risk profile of that transaction. Low-risk transactions—for example, a routine balance enquiry or a small payment to a previously transacted beneficiary—would proceed with standard authentication. High-risk transactions—such as the addition of a new beneficiary, a large fund transfer, or a login from an unrecognised device—would trigger additional authentication challenges, such as a biometric verification, a video-based verification, or a call-back from the bank's contact centre. This approach balances security with customer experience, applying friction selectively where it is most needed rather than uniformly across all transactions.

6.1.7 Strengthen Post-Fraud Victim Support and Rapid Resolution

The experience of fraud victims is a critical but often overlooked dimension of fraud management. Customers who experience fraud and receive a slow, opaque, or bureaucratic response are likely to lose confidence in digital banking and return to cash-based transactions, undermining the bank's financial inclusion objectives. SBI should establish a dedicated fraud victim support function—staffed by trained, empathetic specialists—to provide prompt acknowledgement, clear communication, and transparent case tracking to affected customers. Provisional crediting of disputed amounts within 24 to 48 hours of a verified report, in line with RBI guidelines, should become standard practice. Streamlined resolution processes, supported by AI-assisted case management tools, can significantly reduce resolution times without compromising investigation quality.

6.2 Limitations of the Study

While this study has generated substantive and, it is hoped, useful insights into SBI's fraud detection and prevention framework, several inherent limitations must be acknowledged in the interests of intellectual honesty and to guide the interpretation of findings.

- **Reliance on Secondary Data:** The most significant limitation is the exclusive reliance on publicly available secondary data. SBI's internal fraud detection algorithms, proprietary risk models, actual detection accuracy rates, false positive and false negative rates, and operational performance data are not publicly disclosed. The analysis is therefore necessarily based on what SBI and its regulators have chosen to make public, which may not fully represent the actual state of the bank's fraud detection capabilities. Access to confidential operational data would enable a significantly more precise and empirically rigorous assessment.
- **Dynamic Nature of the Fraud Landscape:** The fraud environment evolves continuously and rapidly. Findings and recommendations that are well-grounded in evidence as of the study period (2020-2024) may require revision as new fraud typologies emerge, as regulatory frameworks are updated, and as the bank's own capabilities develop. The study represents a snapshot of a rapidly moving target and should be read with this temporal limitation in mind.
- **Limited Generalisability:** The findings of this study are specific to SBI Bank and should not be uncritically transferred to other banking institutions, even within the Indian public sector banking context. Different banks have different customer profiles, technology infrastructures, organisational cultures, and resource bases, all of which materially affect fraud detection capabilities and challenges.
- **Absence of Primary Data from SBI Officials:** This study would have been strengthened by direct interviews with SBI's fraud risk management professionals, cybersecurity

specialists, and customer experience teams. The absence of such primary data means that practitioner perspectives are incorporated only through publicly available statements and commentary, which may not fully capture the nuances and complexities of day-to-day fraud management operations.

- **Quantitative Measurement Constraints:** The study was unable to conduct rigorous quantitative benchmarking of SBI's fraud detection performance due to the absence of publicly available data on key performance indicators such as detection rates, false positive ratios, average resolution times, and customer satisfaction scores related to fraud management. Such benchmarking would be a valuable complement to the qualitative analysis provided.

Chapter 7: Bibliography / References

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Appendix: Abbreviations and Glossary

Abbreviation / Term	Full Form / Definition
ACFE	Association of Certified Fraud Examiners
AI	Artificial Intelligence – the simulation of human intelligence by computer systems
ATM	Automated Teller Machine
BHIM SBI Pay	Bharat Interface for Money – SBI's UPI-based payment application
CERT-In	Computer Emergency Response Team – India
CoE	Centre of Excellence
CRILC	Central Repository of Information on Large Credits
CSOC	Cyber Security Operations Centre
EWS	Early Warning System
FS-ISAC	Financial Services Information Sharing and Analysis Center
IBA	Indian Banks' Association
INB	Internet Banking Services (SBI's portal)
LSTM	Long Short-Term Memory – a type of recurrent neural network used in deep learning
MeitY	Ministry of Electronics and Information Technology
MFA	Multi-Factor Authentication
ML	Machine Learning
NACH	National Automated Clearing House
NLP	Natural Language Processing
NPCI	National Payments Corporation of India
OTP	One-Time Password
RBI	Reserve Bank of India
SBI	State Bank of India
SIM	Subscriber Identity Module
TPIN	Transaction Personal Identification Number
UPI	Unified Payments Interface
YONO	You Only Need One – SBI's integrated digital banking platform

Abbreviation / Term	Full Form / Definition
Phishing	Use of deceptive digital communications to trick users into revealing sensitive information
Vishing	Voice phishing – fraud conducted via telephone calls impersonating trusted entities
SIM Swap	Fraudulent transfer of a mobile number to a new SIM card controlled by an attacker
Zero-Trust	Security framework requiring continuous verification of all users, devices, and connections
Behavioural Biometrics	Passive authentication based on patterns of user interaction such as typing rhythm and swipe patterns
Synthetic Identity Fraud	Creation of fictitious identities by combining genuine and fabricated personal data